HS Data Science 26

Session 01: Orientation, Topics, and Workflow

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23 March 2026

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Seminar Data Science

Focus of This Year's Seminar

- retrieval-augmented generation
- agentic search
- open web retrieval on our own infrastructure

Core Idea

This seminar rewards students who can formulate an interesting question, design a manageable study, and defend their choices clearly.

Speaker notes

Open with the change in seminar logic: fewer predefined topics, stronger emphasis on originality, question quality, and communication.

Seminar Topic

The seminar focuses on retrieval-augmented generation and smart agents in the context of open web search.

Technical Context

- we run our own search engine on the Open Web Index
- students can explore topics through this infrastructure
- the environment supports retrieval experiments and agent workflows

Research Perspective

- search is part of the research object
- retrieval choices can be inspected and compared
- the seminar goes beyond a generic RAG demo

Speaker notes

Position the seminar technically: shift from generic ML topics to a coherent research environment built on open retrieval and agentic workflows.

Why OWI and Our Search Engine?

The OWI-based environment exists because web retrieval research is otherwise structurally constrained by proprietary search engines and black-box APIs.

What This Enables

- inspect retrieval behaviour
- compare sparse, dense, and hybrid retrieval
- study provenance and ranking choices
- build agentic workflows on transparent search

Consequence for the Seminar

- the search engine is not just a utility
- the retrieval setup itself can be analysed
- good topics exploit the openness of the system

Demo Search

Use the OURRS demo web search for demonstrations and experiments.

Data and Statistics

- Open Web Index

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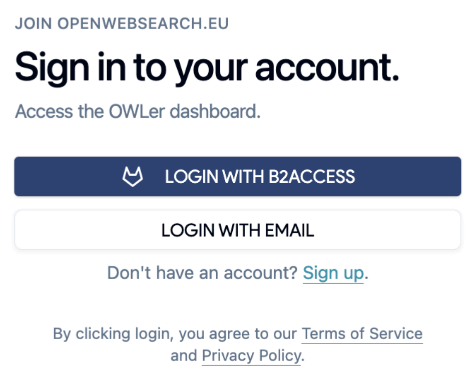
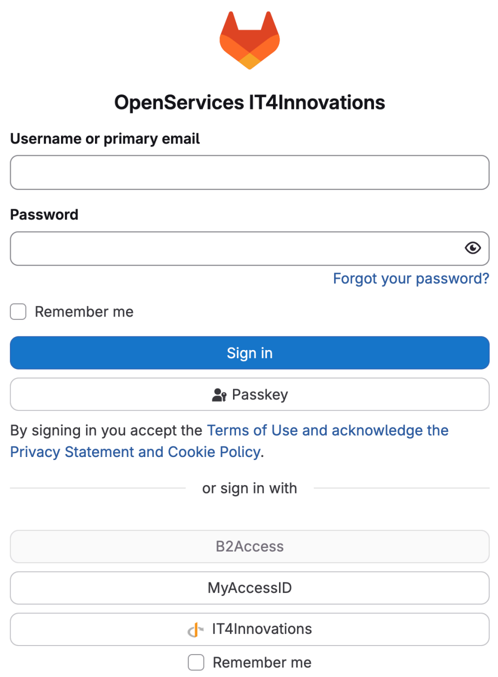
API & Data Download

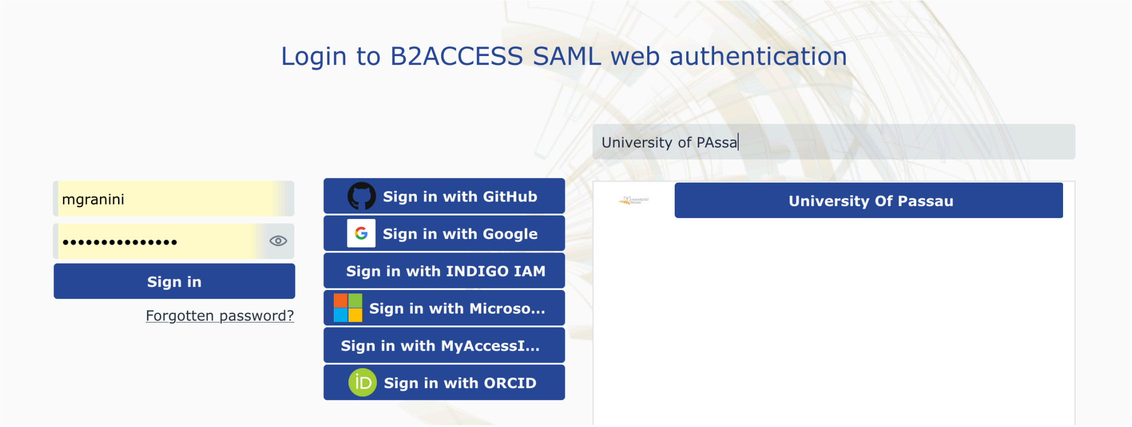
Direct dataL Corpora that could be used: [Open Web Search Curlie 2025]

API Access - Developer Guide:

- https://ourrs.eu/developers
- Login into dashboard: use B2ACCESS and your University of Passau Credentials







How Topic Selection Works

You are expected to generate your own research direction within the seminar theme.

Good Starting Angles

- retrieval quality
- agent planning and verification
- trust and usability
- corpus and query analysis

What Makes a Good Topic

- interesting research question
- manageable scope
- clear evaluation strategy
- relevance to the seminar setup

Example

- weak: *"Build a RAG demo"*
- better: *"When does query reformulation improve agentic search on exploratory tasks?"*

Expected Seminar Work

Scientific Working

- conduct paper research
- identify the gap or tension
- formulate research questions
- document assumptions and limits

Methodology

- implement a prototype or workflow
- run small-scale experiments
- compare alternatives where useful
- explain why the evidence supports the claim

> A modest experiment with a sharp question and clean reasoning is better than an oversized prototype without a defensible result.

Grading

Component	Weight
Independent idea generation	20%
Small-scale but rigorous experiments	15%
Clear methodology and evaluation	15%
Presentation	40%
Report	10%

The asymmetry is intentional: idea generation and presentation are the parts students cannot outsource blindly to AI.

Deliverables / Expected Outputs

All artefacts must be uploaded to the corresponding folders in Stud.IP.

- **Artefact 1 — Research Hypothesis / Seminar Narrative** — 2 pages, ACM format — graded
- **Artefact 2 — Final Report** — 4 pages excl. references, ACM format — graded — deadline 10 September 2026
- **Artefact 3 — Seminar Presentation** — 20 min talk + 10 min Q&A — graded
- a Git repository with code, prompts, experiments, or analysis artefacts

Artefact 1: Research Hypothesis / Seminar Narrative - 12th May 2026

2 pages, ACM two-column format. This artefact is graded.

It serves a dual purpose: it establishes the research agenda up front, and it becomes the benchmark against which the final report is evaluated. The final report will be compared against the promises made here — deviations must be explicitly justified.

The document must contain the following four sections:

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Motivation (1–2 paragraphs) Describe the problem area and its relevance. Situate the work in the current state of the art: what exists, what is missing, and why it matters now.

Research Questions / Research Gap (≥ 3 questions) State at least three concrete research questions. Each question must be accompanied by a brief justification that explains the gap it addresses and why the question is answerable within the seminar scope.

Methodology (≈ ½ page) Describe the methodological approach: which evaluation measures will be used, which datasets or benchmarks will be used, and which tools or frameworks will be applied. The Git repository URL must appear here — it will be checked against the implementation plan throughout the semester.

Timeline / Implementation Plan (1 paragraph, 2–3 milestones) State a concrete plan with 2–3 milestones and target dates. The Git repository commit history will be compared against this plan.

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Deliverables / Expected Outputs - 10th September 2026

Artefact 2: Final Report

4 pages excluding references, ACM two-column format. This artefact is graded. Deadline: 10 September 2026.

Use the ACM Primary Article Template (LaTeX or Word). Upload to the designated Stud.IP folder before the deadline.

The report is a concise scientific paper describing your findings. It will be evaluated against the promises made in Artefact 1 — deviations from the original research questions or methodology must be explicitly justified in the paper.

References — quality matters:

- Use peer-reviewed, published work (conference papers, journal articles) wherever possible.
- An arXiv preprint that has also been formally published at a venue must be cited as the published version. Citing only the arXiv version of a published paper counts **negatively**.
- **Hallucinated references result in automatic failure** — verify every citation before submission.

Deliverables / Expected Outputs - End of June / Beginning of July 2026

Artefact 3: Seminar Presentation

20 minutes presentation + 10 minutes Q&A. This artefact is graded.

The talk should stand alone: a non-expert in your specific topic should be able to follow the motivation, the method, and the main finding. Slides must be uploaded to the designated Stud.IP folder before your presentation date.

The Q&A is part of the assessment — you are expected to answer questions about your methodology, results, and choices made during the project.

Practical Rule

Keep every artefact aligned with one central claim: what question you asked, how you investigated it, and what you learned.

AI Usage

- **Opt-In**: you can use AI for everything, but you have to make it transparent.
- If you use AI, expectations on results, code, report and presentation will be **significantly higher**
- **You fail**
  - if you use AI, but do not make it transparent.
  - if your AI did something wrong (even only a tiny thing) and you did not recognize it.
    - A wrong citation
    - a wrong experiment
    - a wrong claim
    - etc.
  - If your AI did something, and you have
    - no idea what
    - cannot explain concepts, methods, results
    - cannot explain why it did what it did

Organization

Study Programmes

- BSc Computer Science
- BSc Internet Computing
- MSc Computer Science
- MSc AI Engineering
- open to other faculties

Semester Rhythm

- orientation and topic exploration
- question development
- prototype implementation
- presentations with feedback
- refinement and final report

> Lock the question early, then iterate on evidence and presentation quality.

Tentative Schedule

Date	Topic	Your Task
Tue 14/04/26	Topic Presentation, Research Hypothesis	
Tue 21/04/26	Scientific Working I	
Tue 28/04/26	No Seminar	
Tue 05/05/26	Scientific Working II	
Tue 12/05/26	Research Hypothesis Presentation	send narrative;work in feedback; upload accepted narrative
Tue 19/05 – 23/06	Q&A sessions	
Tue 30/06 – 14/07	Student Presentations	Presentation
26/09/10	Submission of final report and code	Submission per E-Mail

All sessions: 10:00–12:00, ITZ R 138.

Next Steps

Before the first week of semester, students should:

- research the field of RAG, agentic search, and web search
- write down one page with a possible topic or research direction
- submit it or present it in the first week of semester

Research-hypothesis presentations are 5–10 minutes.

Spots will be assigned based on the quality of the proposal.

Looking Ahead

Before the next milestone, each student should prepare:

- three candidate topic ideas
- one tentative main research question
- one plausible evaluation strategy
- one short argument for why the question is interesting

> This seminar is about producing an interesting, defensible piece of research in an AI-rich environment where originality, judgment, and communication matter more than routine implementation.